

Customer Behavior Prediction Platform

Final report: Vantara Retail Solutions

THE RESULTS IN THIS REPORT WERE GENERATED ON A SYNTHETIC STAND-IN DATASET (same schema as UCI Online Retail II). THEY ARE NOT REAL-DATA RESULTS. Re-run `python -m src.pipeline` on the real dataset to regenerate this report.

Data source: SYNTHETIC STAND-IN (not real data). Run: run_20261003_092511. Trained: 2026-10-03T09:29:50+00:00. Seed: 42. Training cutoff 2011-09-10 00:00:00; scoring cutoff 2011-12-09 00:00:00.

1. Executive summary

The platform predicts, per customer, the probability of no purchase in the next 90 days (churn), 90-day customer lifetime value, anomalous spending and the most likely next purchase categories, and groups customers into behavioural segments. The recommended production churn model is lightgbm (held-out test ROC-AUC 0.868, recall 0.884, precision 0.794). 6000 customers were scored; 3867 are in the High risk tier.

Success metrics (PRD Section 3.2)

Metric (PRD 3.2)	Target	Observed	Status
Churn ROC-AUC (test)	≥ 0.80	0.868	MET
CLV R^2 (test)	≥ 0.60	0.116	NOT MET
Churn recall (test)	≥ 0.70	0.884	MET
API p95 latency (single prediction)	< 400 ms	44.6 ms	MET
Dashboard load (segment view)	< 3 s	1.40 s	MET
Single-command reproducible pipeline	all steps scripted	<code>python -m src.pipeline</code>	MET

Discussion, CLV R^2 (test): observed 0.116 vs target ≥ 0.60 . Reported as measured, not tuned toward the target. The future-spend target is heavy-tailed and dominated by a few very large customers, which makes R^2 unstable. Model selection used training-set CV R^2 only and chose ridge (xgb_regressor reached test R^2 0.321, but the test set was deliberately not used to pick a model). Candidate remedies to evaluate on the real data: a log-transformed target, a two-stage (purchase probability x spend) model, winsorising extreme customers, or a longer CLV horizon. Class imbalance: accuracy is reported but de-prioritised; precision, recall, F1 and ROC-AUC are always shown together.

2. Data and cleaning

Both yearly sheets are concatenated and ordered chronologically. Cleaning rules: exact duplicates removed; rows without Customer ID flagged (kept for product-level analysis, excluded from customer models); cancellations and negative quantities flagged as returns (not dropped; they feed return_rate); zero/negative prices flagged as adjustments; non-product StockCodes flagged; IQR outliers split into data-entry errors (excluded) and legitimate bulk purchases (kept); descriptions standardised through a StockCode lookup.

Item	Rows / count
rows_in	646,649
exact_duplicates_removed	1,922
rows_missing_customer_id	115,919
return_lines	14,601
price_adjustment_lines	1,281
admin_code_lines	2,563
rows_used_for_customer_model	528,802

Item	Rows / count
unique_customers	6,000

Outliers: 29343 flagged; 6 data-entry errors excluded; 29337 bulk purchases kept.

3. Feature engineering

All features use only transactions strictly before the cutoff date; labels use the following 90 days. Leakage is guarded by an assertion in the builder and by unit tests. Country is one-hot encoded; StockCode is frequency-encoded into avg_product_popularity. Product-category affinity is the share of spend per keyword-derived category. discount_sensitivity is a proxy (the dataset has no discount field): share of spend on lines priced at least 10% below the SKU median.

VIF pruning (threshold 10.0) removed: total_spend, avg_spend. 26 features remain in the model matrix.

4. Churn model comparison

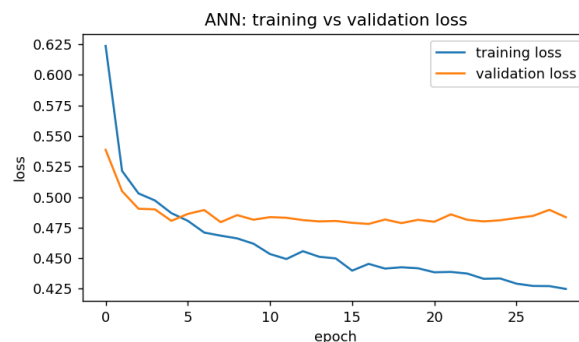
Every model uses the same stratified 70/15/15 split and seed. Hyper-parameters were tuned with 5-fold stratified CV on the training set only; early stopping used the validation split; the operating threshold (highest F1 with recall ≥ 0.70) was chosen on the validation set; the test set was used once for the numbers below. Ranked by ROC-AUC then recall.

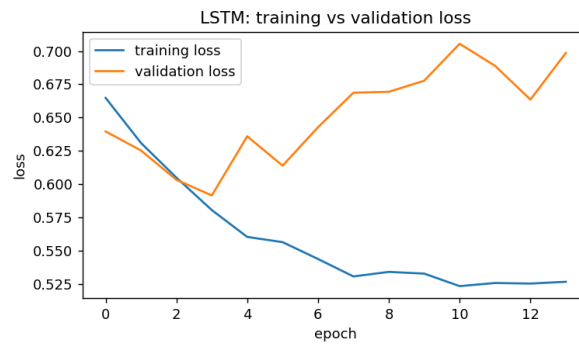
#	Model	CV AUC	Test AUC	Recall	Precision	F1	Accuracy
1	LightGBM	0.866	0.868	0.884	0.794	0.836	0.791
2	XGBoost	0.87	0.865	0.907	0.771	0.833	0.781
3	Random Forest	0.871	0.864	0.89	0.788	0.836	0.788
4	Logistic Regression	0.865	0.863	0.867	0.797	0.831	0.786
5	ANN (feed-forward)	0.861	0.86	0.907	0.765	0.83	0.775
6	Decision Tree	0.855	0.855	0.822	0.835	0.829	0.794
7	KNN	0.823	0.802	0.865	0.725	0.789	0.719
8	LSTM (sequence)	0.84	0.799	0.888	0.727	0.8	0.731

Confusion matrix, lightgbm, test set (threshold 0.318)

	Predicted retained	Predicted churned
Actual retained	226	123
Actual churned	62	473

Recommendation. lightgbm is recommended for the production API. It was selected among the tree-based candidates by validation ROC-AUC (recall as tie-break). Tree ensembles are exactly compatible with SHAP TreeExplainer, score a single customer in milliseconds, need no feature scaling and, with only about 5890 customers, are the safer choice than deep networks (PRD Section 19). The top-ranked model overall by test AUC is LightGBM (0.868); where it differs from lightgbm, the gap is small compared with the explainability and latency benefits. The ANN and LSTM are retained as comparative models.





5. CLV, next category and anomaly detection

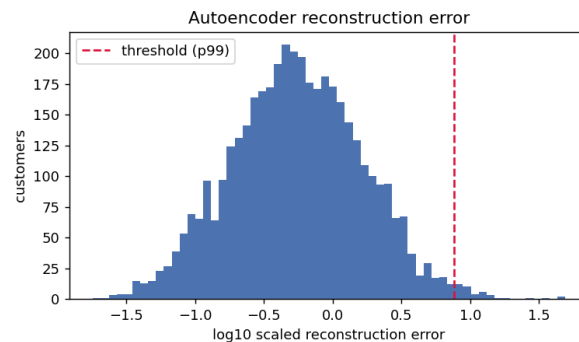
Model	MAE	RMSE	R ² (test)	CV R ² (train)
ridge	336.16	697.12	0.116	0.175
xgb_regressor	268.06	610.86	0.321	-0.049

Selected CLV model (by training-set CV R²): ridge. Target: spend in the 90 days after the cutoff.

Next purchase category model (multiclass random forest): accuracy 0.542, macro-F1 0.537.

Autoencoder anomaly detector: threshold at the 99th percentile of training reconstruction error (7.5919); flag rate train/val/test = 1.019% / 1.471% / 1.471% (expected 1.00%), so the threshold transfers to held-out customers.

Independent check of the flags: Spearman rank correlation of the score with Mahalanobis distance 0.32 and with an Isolation Forest 0.21; share of flagged test customers inside the Isolation Forest top 5%: 0.07692307692307693. Weak agreement means the flags should be treated as a review list, not as confirmed anomalies.



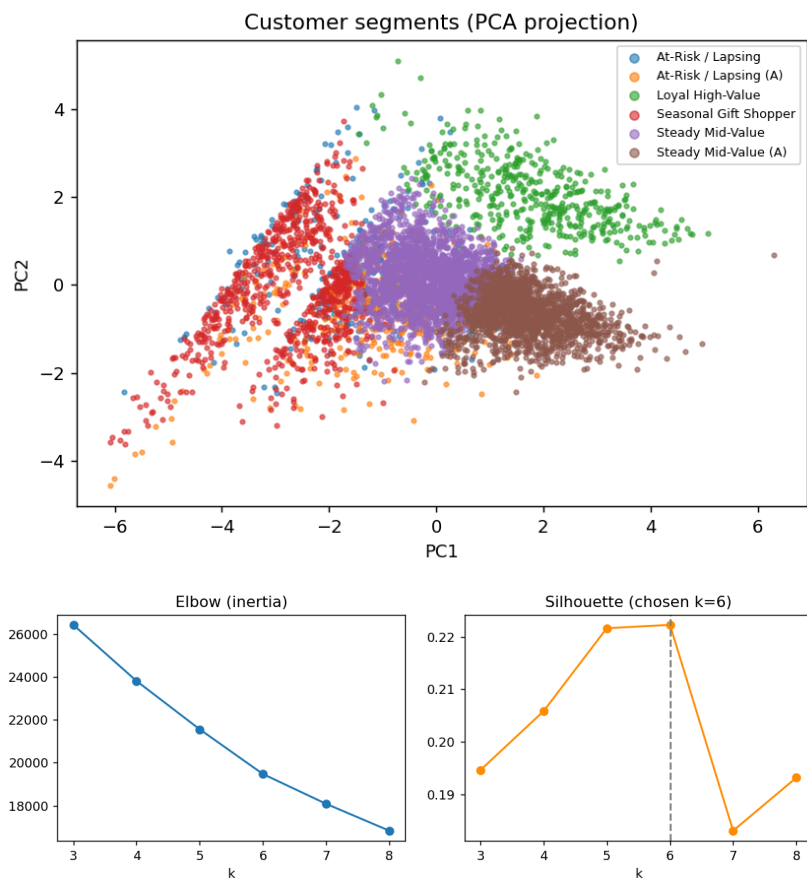
6. Customer segmentation

K-Means with k=6 chosen by silhouette over k=3..8 (elbow plot reviewed): silhouette 0.222,

Davies-Bouldin 1.280. Secondary algorithm (GMM): silhouette 0.05107804288914508, Davies-Bouldin

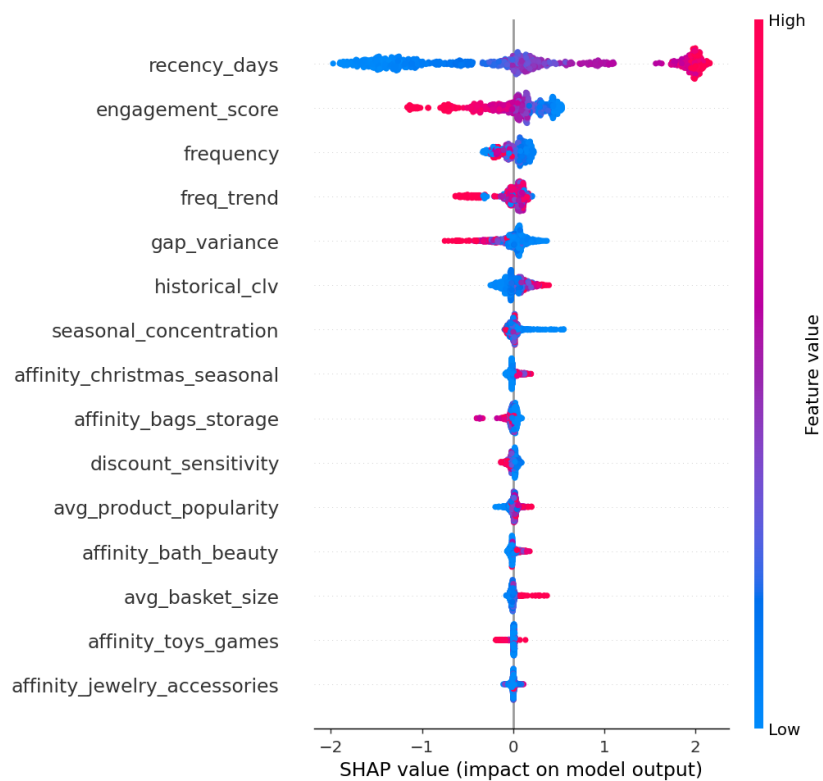
2.421207126138935, adjusted Rand vs K-Means 0.2321057522671958.

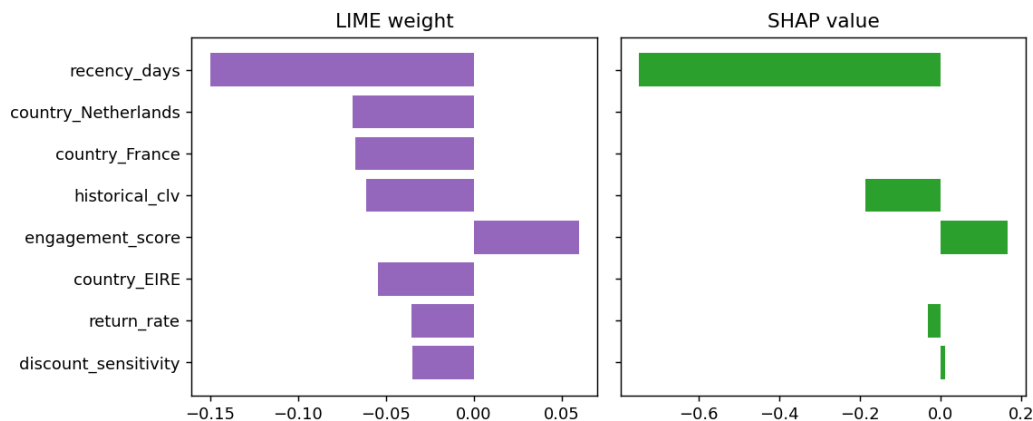
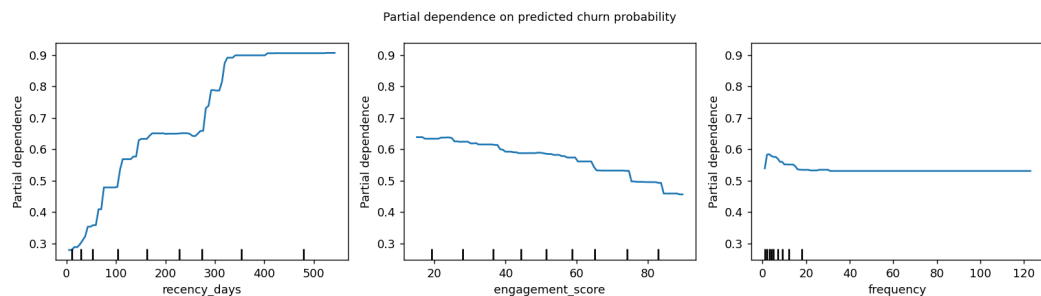
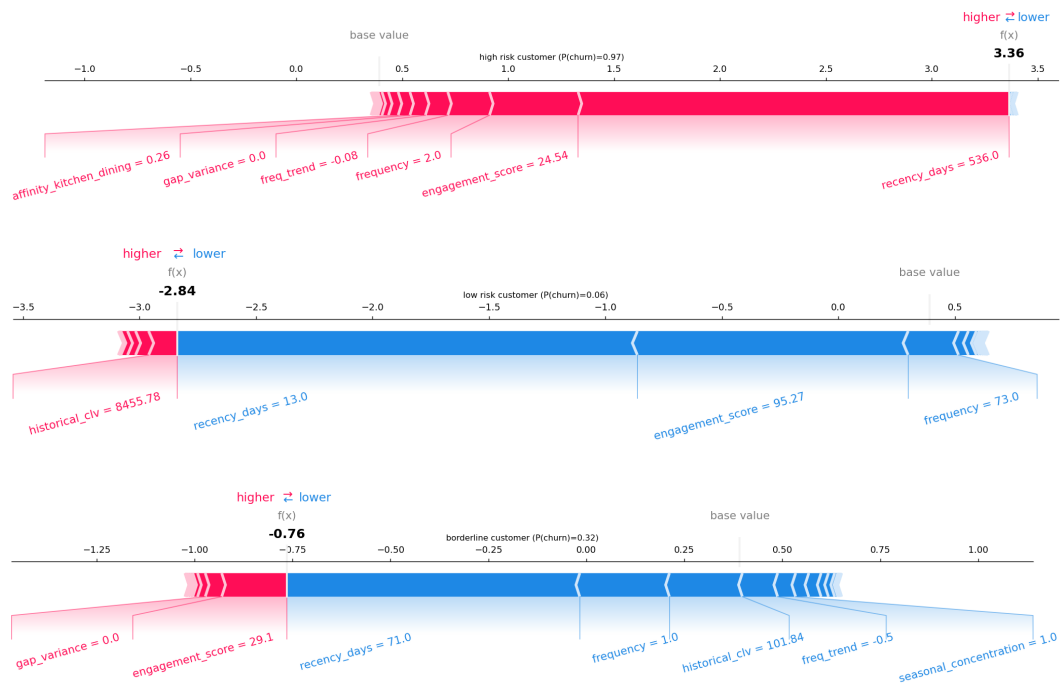
segment_label	customers	share_pct	recency_days	frequency	total_spend	return_rate	mean_churn_probability
At-Risk / Lapsing (A)	512	8.5	297.3	4.71	547.67	0.02	0.69
Loyal High-Value	456	7.6	192.22	11.21	7860.9	0.02	0.53
Steady Mid-Value (A)	1638	27.3	39.57	20.22	2648.5	0.03	0.21
Steady Mid-Value	2211	36.8	270.86	6.2	828.91	0.02	0.7
Seasonal Gift Shopper	818	13.6	412.48	1.92	174.24	0.0	0.84
At-Risk / Lapsing	365	6.1	330.79	3.52	455.32	0.12	0.74



7. Explainability

Global SHAP importance, three representative customers, partial dependence of the top features and a LIME-vs-SHAP comparison are below.





LIME vs SHAP for the borderline customer: top-5 overlap 60% (engagement_score, historical_clv, recency_days), sign agreement 0.7142857142857143. Differences are expected: SHAP attributes the model's own output exactly, LIME fits a local linear surrogate on perturbed samples.

Plain-language template example: This customer is flagged as HIGH churn risk: predicted churn probability 32%, at or above the 32% operating threshold. Main drivers (relative to the average customer): days since last purchase = 71.00 (lower than typical) lowers the risk; number of orders = 1.00 (lower than typical) lowers the risk; net lifetime revenue = 101.84 (lower than typical) lowers the risk.

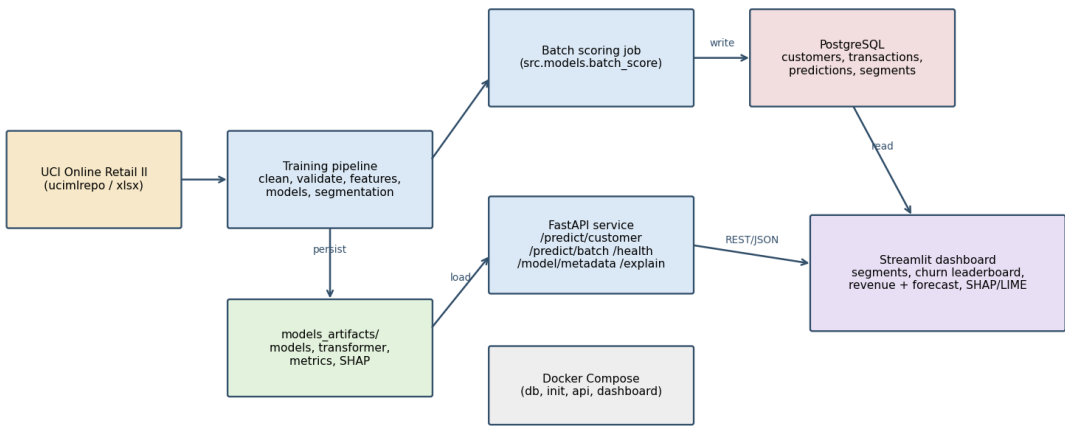
8. Serving, deployment and measurements

FastAPI exposes /predict/customer, /predict/batch (CSV), /model/metadata, /health and /explain/customer; Pydantic models validate every record. Predictions and segments are persisted in PostgreSQL; the Streamlit dashboard reads them and calls the API for batch scoring and LIME. Everything starts with docker-compose up.

API latency (200 requests): p50 42.3 ms, p95 44.6 ms, p99 46.9 ms. Method: in-process TestClient, single worker (excludes network). Dashboard server-side run: first 1.40 s, cached 0.36 s (Streamlit AppTest server-side script run)

(excludes browser rendering)). Both were measured in a 1-CPU sandbox, not on production hardware or over a network.

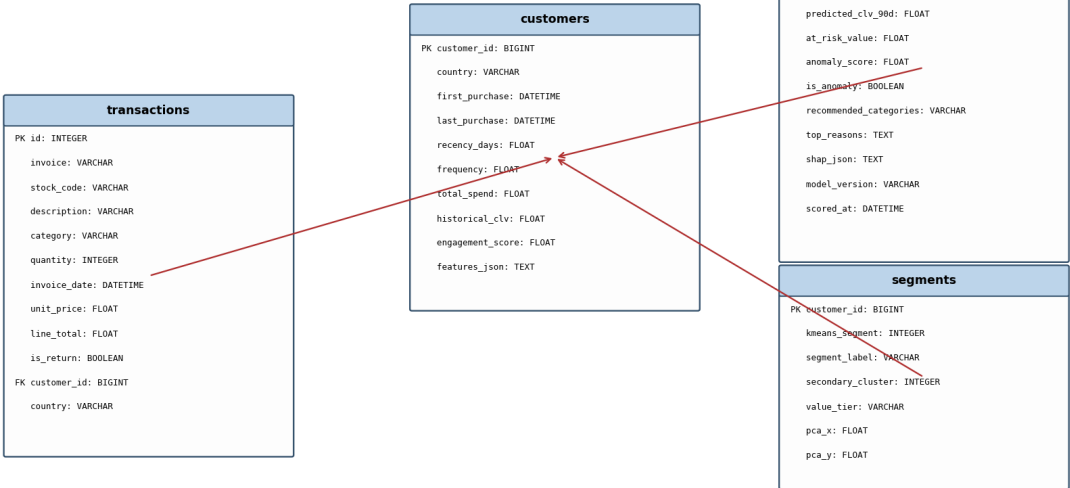
System architecture



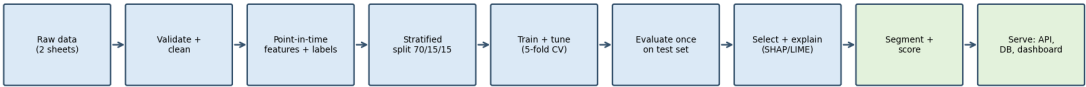
predictions

Entity-relationship diagram

arrows: foreign key -> referenced table (many-to-one)



ML workflow



Test set is used once for reporting. Class imbalance is handled on training data only. Scalers/encoders are fitted on training data and persisted.

Appendix A. Mathematical foundations

A.1 Statistics and probability

RFM and the engagement score use percentile ranks: $\text{score} = 100 \cdot (1 - \text{PR}(\text{recency}) + \text{PR}(\text{frequency}) + \text{PR}(\text{monetary}))/3$, where $\text{PR}(x)$ is the empirical CDF value from the fitted reference distribution. Next-purchase probability is $P(\text{purchase} \mid \text{recency, frequency, monetary, ...}) = 1 - \hat{y}$, the classifier's predicted probability.

A.2 Linear algebra and PCA

The feature matrix is $X \in \mathbb{R}^{(n \times d)}$. PCA diagonalises the covariance $\Sigma = (1/n)X^T X$ (X centred): $\Sigma v = \lambda v$. The segment plots project onto the two eigenvectors with largest λ (explained variance: 43.2%, 16.1%). K-Means minimises $\sum_k \sum_{x \in C_k} \|x - \mu_k\|^2$; silhouette $s = (b - a)/\max(a, b)$.

A.3 Logistic regression and cross-entropy

$P(y=1|x) = 1/(1 + e^{-(w^T x + b)})$; loss $L = -(1/n) \sum [y_i \log \hat{y}_i + (1-y_i) \log(1-\hat{y}_i)]$. Class weights multiply each term by w_c .

A.4 Gradient descent and backpropagation

$w \leftarrow w - \eta \nabla L(w)$. Backpropagation applies the chain rule layer by layer: $\partial L / \partial W^{(l)} = \delta^{(l)} (a^{(l-1)})^T$ with $\delta^{(l)} = ((W^{(l+1)})^T \delta^{(l+1)}) \odot f'(z^{(l)})$. Adam rescales the step by running moment estimates. For the LSTM, gradients flow back through time via the gate equations $c_t = f_t \odot c_{t-1} + i_t \odot \tilde{c}_t$. The training-vs-validation loss curves (Section 4) confirm convergence and flag overfitting; early stopping restores the best validation weights.

A.5 Regularisation

L1: $L + \lambda \|w\|_1$ (sparse weights); L2: $L + \lambda \|w\|_2^2$ (shrinkage). In this code, logistic regression tunes $C = 1/\lambda$ and $\text{l1_ratio} \in \{0, 1\}$; the ANN uses an L2 kernel penalty. Dropout multiplies activations by Bernoulli($1-p$) masks during training, an implicit ensemble average.

A.6 Evaluation metrics

Precision = $TP/(TP+FP)$; Recall = $TP/(TP+FN)$; $F1 = 2PR/(P+R)$; ROC-AUC = $P(\text{score of a random churner} > \text{score of a random retained customer})$, the area under TPR vs FPR over all thresholds. MAE = $\text{mean}|y-\hat{y}|$, RMSE = $\sqrt{\text{mean}(y-\hat{y})^2}$, $R^2 = 1 - \Sigma(y-\hat{y})^2/\Sigma(y-\bar{y})^2$. Davies-Bouldin = mean over clusters of $\max_{j \neq i} (\sigma_i + \sigma_j)/d(\mu_i, \mu_j)$ (lower is better).

A.7 Tree ensembles and gradient boosting

A decision tree chooses the split that most reduces impurity; for the Gini index $G = 1 - \sum_k p_k^2$, the gain of splitting a node into children L and R is $\Delta G = G - (n_L/n)G_L - (n_R/n)G_R$. Random forests average B trees grown on bootstrap samples with random feature subsets, which reduces variance: $\text{Var}(\text{mean}) = p\sigma^2 + (1-p)\sigma^2/B$ for pairwise correlation p . Gradient boosting builds $F_m(x) = F_{m-1}(x) + v \cdot h_m(x)$, where h_m fits

the negative gradient $r_i = -\partial L(y_i, F(x_i))/\partial F(x_i)$ (for log-loss $r_i = y_i - \hat{p}_i$). XGBoost adds a second-order Taylor expansion and a complexity penalty $\gamma T + \frac{1}{2}\lambda \|w\|^2$ per tree; `scale_pos_weight` rescales positive-class gradients to handle imbalance. Early stopping halts adding trees when the validation AUC stops improving.

A.8 Sequence model and autoencoder

The LSTM cell updates its state with $f_t = \sigma(W_f[h_{t-1}, x_t] + b_f)$, $i_t = \sigma(W_i[h_{t-1}, x_t] + b_i)$, $\tilde{c}_t = \tanh(W_c[h_{t-1}, x_t] + b_c)$, $c_t = f_t \odot c_{t-1} + i_t \odot \tilde{c}_t$, $o_t = \sigma(W_o[h_{t-1}, x_t] + b_o)$, $h_t = o_t \odot \tanh(c_t)$. Each input step is [amount, gap in days, is-event] plus a category embedding; a final step carries the days from the last order to the prediction date. The autoencoder learns $\hat{x} = g(f(x))$ by minimising $(1/n)\sum \|x - \hat{x}\|^2$; the anomaly score here is $(1/d)\sum_j (x_j - \hat{x}_j)^2/\sigma_j^2$ with σ_j^2 the training residual variance of feature j , flagged above the 99th training percentile.

A.9 Operating threshold and class weights

With class weights w_0, w_1 the loss becomes $-(1/n)\sum w_{y_i}[y_i \log \hat{y}_i + (1-y_i)\log(1-\hat{y}_i)]$, where $w_c = n/(2 \cdot n_c)$ for 'balanced'. The decision rule is $\hat{y} = 1[\hat{p} \geq \tau]$; τ is chosen on the validation set as the F1-maximising threshold among those with recall ≥ 0.70 , then frozen before the test set is scored.

A.10 SHAP

Shapley values $\phi_i = \sum_{S \subseteq F \setminus \{i\}} \frac{|S|!(|F|-|S|-1)!}{|F|!} \cdot [f(S \cup \{i\}) - f(S)]$ satisfy local accuracy $f(x) = \phi_0 + \sum \phi_i$. TreeExplainer computes them exactly in polynomial time for tree ensembles; for XGBoost/LightGBM the values are in log-odds space.

Appendix B. Future enhancements

- Transformer-based sequence model for purchase timing (needs a larger event log and more tuning).
- Streaming / near-real-time scoring (event ingestion, incremental feature store).
- API authentication, rate limiting and CI/CD with automated retraining and model registry.
- A/B testing of retention campaigns to measure causal uplift, and drift monitoring on feature and score distributions.
- Additional boosted-tree variants (e.g. CatBoost) and probability calibration.